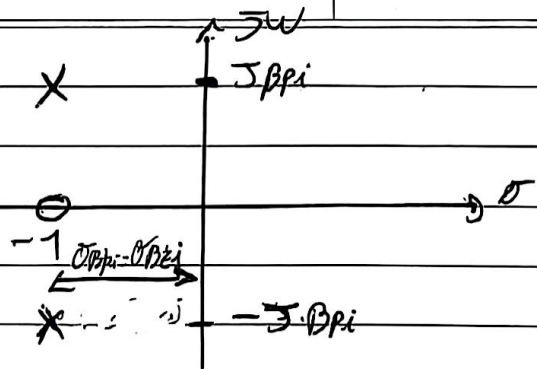


TP3 Ejercicios 1) a)

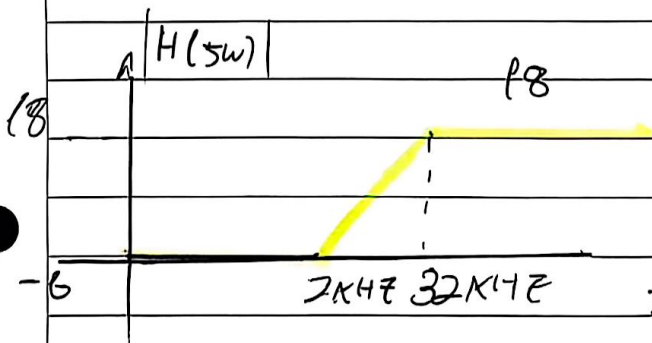
$$H(s) = \frac{s + 1}{s^2 + 2s + 5}$$



$$H(s) = \frac{s + 1}{[s - (-1 \pm 2j)]}$$

Buscamos retardo $\Rightarrow G = \frac{1}{1+(w-2)^2} + \frac{1}{1+(w+2)^2} - \frac{1}{1+w^2}$

Descomponiendo $G = \frac{w^4 + 18w^2 - 15}{w^6 - 5w^4 + 19w^2 + 25}$



$$m = \frac{\Delta \text{[dB]}}{\Delta \text{[dec]}} = \frac{18 - (-6)}{\log_{10} \left[\frac{32 \text{ KHz}}{2 \text{ KHz}} \right]} = \frac{24 \text{ dB}}{1.2 \text{ dec}}$$

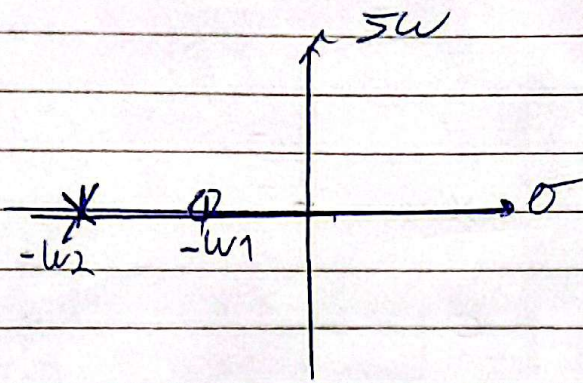
$32 = 2^5$, y 5 rangos de octavas
 $2 = 2^1$, y 1 rango de octavas
 $5 - 1 = 4$ rangos de octavas entre $f_s - f_i$

1 octava es el doble de f de onda, 6dB por octava, 24dB Sabros

$$\Delta X \text{ [dB]} = \log_{10} \left[\frac{f_{sup}}{f_{inf}} \right] = \text{"cont." de octavas}$$

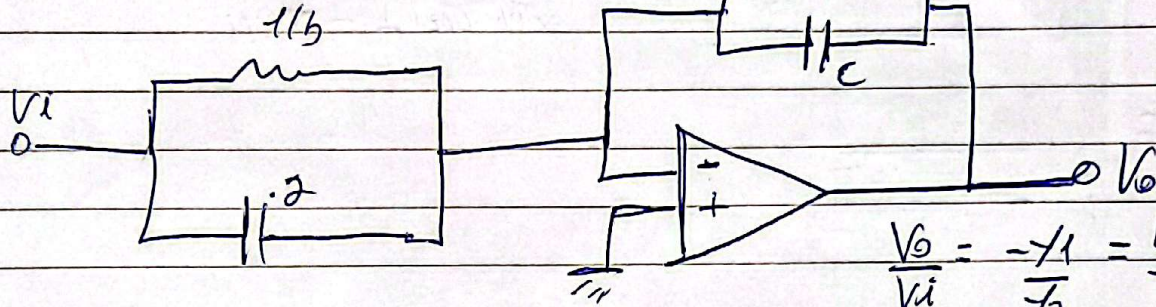
$$m = \frac{24 \text{ dB}}{1.2 \text{ dec}} = \frac{20 \text{ dB}}{\text{dec}} \quad m = \frac{24 \text{ dB}}{4 \text{ oct}} = \frac{6 \text{ dB}}{\text{oct}}$$

F_{inferior} en $w_1 = 2 \text{ KHz} \cdot 2\pi$) El zero viene con 0 dB la respuesta
 Poles en $w_2 = 32 \text{ KHz} \cdot 2\pi$ y luego viene con 20 dB la respuesta
 sample dec entre 2K y 32K



$$H(s) = K \cdot \frac{s + \omega_1}{s + \omega_2}$$

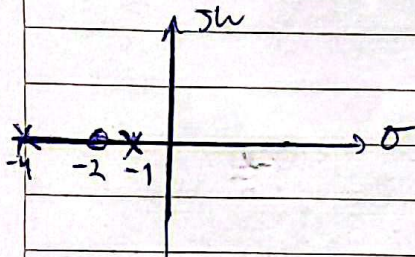
↓ Balanced
no dc gain
1/d



$$\frac{V_o}{V_i} = -\gamma_1 = \frac{b + s \cdot 2}{s + 1/c}$$

$$H(s) = \frac{2}{c} \cdot \frac{s + b/2}{s + 1/c}$$

1) b) $H(s) = \frac{s+2}{(s+1)(s+4)}$, *plantas desgranado de polos y ceros*



A partir del diagrama obtengo el retardo

$$\Rightarrow G = \sum \frac{\sigma_{p_i}}{\sigma_{p_i}^2 + (w \pm \beta_{p_i})^2} - \sum \frac{\sigma_{z_i}}{\sigma_{z_i}^2 + (w \pm \beta_{z_i})^2}$$

$$G = \frac{1}{1^2 + (w \pm 0)^2} + \frac{4}{4^2 + (w \pm 0)^2} - \frac{2}{2^2 + (w \pm 0)^2}$$

$$G = \frac{1}{1 + w^2} + \frac{4}{16 + w^2} - \frac{2}{4 + w^2}$$

$$G = \frac{(16 + w^2) \cdot (4 + w^2) + 4(1 + w^2) \cdot (4 + w^2) - 2(1 + w^2)(16 + w^2)}{(1 + w^2)(16 + w^2)(4 + w^2)}$$

$$G = \frac{(64 + 16w^2 + w^2 \cdot 4 + w^4) + 4(4 + w^2 + w^2 \cdot 4 + w^4) - 2(16 + w^2 + w^2 \cdot 16 + w^4)}{(16 + w^2 + w^2 \cdot 16 + w^4)(4 + w^2)}$$

$$G = \frac{w^4 + 20w^2 + 64 + 4w^4 + 20w^2 + 16 - 2w^4 - 34w^2 - 32}{(w^4 + 17w^2 + 16)(4 + w^2)}$$

$$G = \frac{3w^4 + 6w^2 + 48}{4w^4 + 68w^2 + 64 + w^6 + 17w^4 + 16w^2}$$

$$G = \frac{3w^4 + 6w^2 + 48}{w^6 + 21w^4 + 84w^2 + 64}$$